Theory, Assumption, Hypothesis

What's a Theory? What Do We Assume? Why Hypotheses?

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Where We Left Off

Why do we do literature reviews?

- To avoid "reinventing the wheel"
- To lend credibility to the research question.
- To provide scholarly context to the topic.
- To "carve out" a place for our own contribution within the literature.

The best lit reviews:

- Focus primarily on the dependent variable (no matter the case, IVs, or scope).
- Synthesize the scholarship into patterns among explanations of the DV outcome.
- Evaluate literature on methodological or theoretical grounds.
- Identify a clear and relevant "gap" in the literature.
- Do NOT look like annotated bibliographies.
- Do NOT simply summarize each article.

Why Do We Theorize?

We theorize in order to provide simpler answers to complex problems in a complex world.

- Example puzzle: There are two main parties in U.S. the U.S. political system but there are multiple parties elsewhere, and many people seem to dislike both parties here.
- Example research question: What explains the emergence of two-party systems?
- What are examples of possible theories that explain why we have a two-party system?

What is a Theory

A **theory** is a "reasoned and precise speculation about the answer to a research question, including a statement about why the proposed answer is correct."

Theories help us *describe*, *predict*, and *prescribe*.

So, what makes a theory "good" and scientific?

“Good” Theories

Good, *scientific* theories must be:

- testable (can we test it?).
- falsifiable (is it possible to refute it?).
- logically complete (does it explain the whole phenomenon?).
- logically consistent (is it free of contradictions?).
- empirically supported (can we rely on objectively observable data?).

Important questions to ask about your theory:

- Can you communicate it well?
- Is it generalizable? To what population of cases?
- How parsimonious is it?

***Good* theories don’t have to be *perfect* theories; sometimes things have multiple explanations.**

The Parts of A Theory

1. The **explanation** (i.e., the short answer to the research question).
2. The **mechanism** (i.e., X causes Y, but *how* exactly?).
3. The **assumption** (i.e., implicit or explicit, but must be reasonable).
4. The **conditions** (i.e., What is the scope over which your answer is true?).

Assume = Ass + U + Me?

What needs to be true in order for your theory to be correct?
Why is your causal mechanism reasonable?

You can't simply offer an assumption instead of a theory...

- Assumptions *could* be tested/falsified, but not easily or here.
- Can be functional and temporary (to advance a model in early stages).
- Are often conditional (e.g., if this is true, then...).
- Not usually controversial or arbitrary.

Clear Assumptions are Better Assumptions

If we had to empirically establish everything all the time, nothing would get done!

What do background assumptions look like?

- Determinism is an assumption (that things aren't random)
- Objective reality is an assumption
- The universality of gravity is an assumption
- Rational choice is an assumption

One HypothesIs, Two HypothesEs

A **hypothesis** is specific, testable expectations about reality, based on the argument.

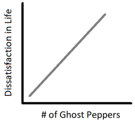
- It's a statement of the relationship that you expect to find between an IV and your DV.

How hypotheses can look:

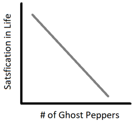
- The higher (lower) the X, the higher (lower) the Y.
- As X increases (or decreases), Y increases (or decreases).
- When X is true (or false), Y is true (or false)
- As X increases (or decreases), the probability of Y being true increases (or decreases).

Some Examples of Relationships

Positive vs. Negative = Direct vs. Indirect

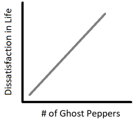


Positive
(Direct)

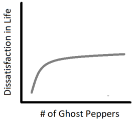


Negative
(Indirect/inverse)

Linear vs. Nonlinear

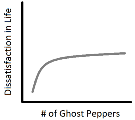


Linear

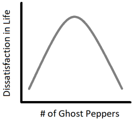


Nonlinear

Monotonic vs Nonmonotonic



Monotonic



Non-monotonic